

Purpose and Primary Function

Introduction Paragraph:

Keep track of all clients that borrow assets from the library, as well as their membership type which affects their borrow limit and fee rate. The client can either reserve or borrow an asset and the database will keep track of which asset they borrowed and what date they did so it can keep track whether it is overdue. When reserving an asset the client will be given a queue time that tracks how long they've reserved an asset for. The asset entity has 3 specializations such as books, magazines, and digital media which all have different attributes.

Scope

Scope Statement: Define the scope of the project: what will be included.

- Store data for library materials, customers, and history.
- Enable creating/updating entries (tuples) for all database tables.
- Enable arbitrary queries and structure default data reports.
- **Intended platform:** Computer
- **Anticipated user base:** developers, customers, library managers, library staff?
- **Preliminary interface ideas:** Admin Login/User Login, Info access, Inventory Search (both), Checkout (admin), Account management

Glossary

- **Entities**
 - **Client**
 - **Attributes:**
 - Unique ID
 - First Name
 - Last Name
 - Email
 - Account Status
 - Borrowing Restrictions
 - Membership Type
 - **Asset**
 - **Attributes:**
 - Item ID
 - Availability Status
 - **Subtypes:**
 - **Book**
 - **Attributes:**
 - Author
 - ISBN
 - Title
 - Publication Date
 - Rarity
 - Genre
 - **Digital Media**
 - **Attributes:**

- Title
 - Creator
 - Genre
 - ISBN
 - Publication Year
- **Magazine**
 - **Attributes:**
 - Rarity
 - Title
 - Issue Number
 - Publication Date
- **Relationships**
 - **Transaction**
 - **Cardinality:**
 - A client may make a transaction on zero or more assets
 - An asset may be borrowed by zero or more clients, but can only be borrowed by one person at a time
 - **Attributes:**
 - Fee Status
 - Borrow Date
 - Transaction Type
 - Renewal Count
 - Due Date
 - Transaction ID
 - Return Date
 - **Reserve**
 - **Cardinality:**
 - A client may make a reservation on zero or more assets
 - An asset may be reserved by zero or more people
 - **Attributes:**
 - Queue Time